ORBIT-INTEL 6G: Intelligent Resource Management and Predictive Reliability Orchestration for Integrated Non-Terrestrial Networks

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Repository: `https://github.com/SlockAhuja/orbit-intel-6g` · Showcase: `https://slockahuja.github.io/orbit-intel-6g/`

Abstract

Integrated Space-Air-Ground Non-Terrestrial Networks (SAGIN) form an indispensable pillar of Sixth-Generation (6G) wireless systems, delivering pervasive connectivity across maritime, aeronautical, and remote wilderness regions. However, high orbital velocity ($v \approx 7.56$ km/s), brief visibility windows (< 8 min), Doppler shifts (± 48 kHz), and atmospheric rain fade (> 15 dB) severely challenge Ultra-Reliable Low-Latency Communication (URLLC). This paper presents **ORBIT-INTEL 6G**, a two-tier intelligence framework addressing these bottlenecks. At Tier 1, an intra-tier Proximal Policy Optimization (PPO) reinforcement learning agent governs dynamic user association, orthogonal subcarrier allocation, and adaptive power control. At Tier 2, Confidence-Aware Predictive Reliability Orchestration (CAPRO) performs multi-horizon link forecasting ($H \in \{1, 3, 5, 10\}$ s) with distribution-free split-conformal calibration ($1 - \alpha = 95\%$) and dual aleatoric-epistemic uncertainty estimation to execute risk-adjusted proactive handovers. Reported experimental results from an extensive 400-trial Monte Carlo benchmark campaign demonstrate a packet delivery ratio of **99.41%**, a 99th-percentile tail latency of **6.82 ms** (meeting sub-10ms URLLC requirements), an outage probability of **0.59%**, and an **89.3%** reduction in handover ping-pong flapping relative to conventional 3GPP reactive baselines.

Keywords—6G, Non-Terrestrial Networks, SAGIN, Resource Management, PPO, Predictive Intelligence, CAPRO.

1. Introduction

Non-Terrestrial Networks (NTN) have emerged as a foundational architecture in 3GPP Releases 17 and 18 to bridge the digital divide by integrating satellite, high-altitude platform, and terrestrial infrastructure into unified 6G communications. The resulting Space-Air-Ground Integrated Network (SAGIN) offers unprecedented coverage resilience across oceanic transport routes, polar corridors, and disaster zones. However, delivering deterministic URLLC guarantees over non-terrestrial links is hindered by three core physical limitations: rapid Low Earth Orbit (LEO) satellite motion inducing transient topocentric visibility windows (< 8 minutes); severe Doppler dynamics (± 48 kHz at Ka-band 28 GHz); and dynamic atmospheric attenuation (ITU-R P.618) during convective precipitation events.

Crucially, standard cellular handover mechanisms (such as 3GPP Event A3/A4 triggers) operate purely reactively, initiating target node search only after signal degradation or radio link failure (RLF) has already manifested. In satellite-terrestrial feeder topologies characterized by round-trip propagation delays of tens of milliseconds, reactive re-association induces packet dropping, retransmission storms, and severe handover ping-pong flapping. To overcome these constraints, 6G NTN requires a paradigm shift toward uncertainty-aware proactive orchestration.

2. Proposed ORBIT-INTEL 6G Framework

The ORBIT-INTEL 6G architecture models a five-tier heterogeneous SAGIN hierarchy: Tier 1 comprises LEO satellites (500–600 km altitude) providing wide-area backhaul and dynamic spot beams; Tier 2 incorporates stratospheric High-Altitude Platform Stations (HAPS, 20 km) acting as quasi-stationary aerial base stations free from tropospheric rain fade; Tier 3 deploys tactical Unmanned Aerial Vehicles (UAVs, 0.3–2 km) for localized line-of-sight restoration in obstructed areas; and Tiers 4 and 5 encompass terrestrial gNodeBs (30 m) and mobile user terminals distributed across five representative geographic environments (Urban Hotspots, Rural Farmlands, Maritime Corridors, Mountain Observatories, and Post-Disaster Emergency Zones). The platform couples a standard-compliant orbital propagation engine with a two-tier intelligence hierarchy: an intra-tier physical resource optimizer and a cross-tier predictive reliability orchestrator.

3. AI-Based Resource Management

Physical-layer radio resource allocation within active satellite footprints is orchestrated by an autonomous deep reinforcement learning agent based on Proximal Policy Optimization (PPO). Interacting with a standardized Gymnasium non-terrestrial environment, the agent observes the normalized state $s_t = [h_t, \gamma_t, B_t, P_t, \text{Alt}_t] \in \mathbb{R}^5$, representing channel gain, SINR, available bandwidth, transmit power budget, and satellite altitude. The actor-critic network selects discrete actions governing: (i) *Terminal Association*, weighting topocentric elevation and dwell time with a hysteresis bonus ($+4.5$) to prevent premature handoffs; (ii) *Orthogonal Subcarrier Slicing*, assigning orthogonal frequency partitions (16–32 channels) across adjacent spot beams to eliminate co-channel interference; and (iii) *Water-Filling Power Control*, allocating transmit power ($P_c \leq P_{\max}$) to mitigate localized path attenuation. The policy optimizes the scalar reward:

$$R_t = \text{Throughput}_t - \text{Delay}_t - 10 \cdot \text{PacketLoss}_t \tag{1}$$

Heavily penalizing packet losses, the PPO agent reliably maintains maximal spectral efficiency during stable orbital passes.

4. Predictive Reliability Orchestration

While physical-layer PPO manages intra-tier spectrum and power slicing, network-layer continuity across constellation handovers requires predictive intelligence. ORBIT-INTEL 6G introduces Confidence-Aware Predictive Reliability Orchestration (CAPRO), operating as an independent network-layer framework. Crucially, PPO and CAPRO perform distinct functions: PPO manages intra-tier physical resources, whereas CAPRO orchestrates cross-tier proactive handovers and routing across LEO, HAPS, and UAV relays. They are complementary components rather than a single joint model.

CAPRO executes three sequential stages: (i) *Multi-Horizon Neural Forecasting*, ingesting rolling telemetry time-series to predict link quality trajectories over lookahead horizons $H \in \{1, 3, 5, 10\}$ s; (ii) *Split-Conformal Calibration*, deriving distribution-free prediction intervals $[\hat{y}^{\text{lower}}_{t+H}, \hat{y}^{\text{upper}}_{t+H}]$ from non-conformity residuals with finite-sample marginal coverage $\mathbb{P}(Y_{t+H} \in \mathcal{C}(X_t)) \geq 95\%$, decomposing variance into aleatoric channel noise σ_{aleat} and epistemic model uncertainty σ_{epist}; and (iii) *Risk-Adjusted Utility Maximization*, ranking candidate target nodes $c \in \{\text{LEO}_2, \text{HAPS}, \text{UAV}\}$ via:

$$\mathcal{U}(c) = \hat{\gamma}_c(t+H) - \lambda \cdot (\sigma_{\text{aleat},c}(t+H) + \sigma_{\text{epist},c}(t+H)) - C_{\text{handover}} \tag{2}$$

where $\lambda = 1.5$ balances risk aversion and C_{handover} penalizes signaling overhead. When the active link utility degrades, CAPRO triggers a proactive handover 1.8 s before line-of-sight failure, enabling seamless transition without packet loss.

5. Simulation and Experimental Methodology

The ORBIT-INTEL 6G evaluation framework implements physical aerospace kinematics and propagation models adhering to international telecommunications standards:

- **Kinematics & Orbit Propagation**: LEO circular orbits at 500–600 km altitude with minimum elevation threshold $\theta \geq 20^\circ$. Slant range distance is evaluated geometrically as $d(t) = \sqrt{R_E^2 \sin^2\theta + 2R_E h + h^2} - R_E \sin\theta$, where $R_E = 6371$ km is Earth's mean radius and h is orbital altitude.
- **Path Loss & Doppler Dynamics**: Free Space Path Loss obeys 3GPP TR 38.811 at Ka-band ($f_c = 28$ GHz): FSPL $= 20\log_{10}(d) + 20\log_{10}(f_c) - 147.55$ dB. Doppler shifts $f_D(t) = f_c \cdot (v_{\text{rel}}(t)/c)$ span ± 48 kHz across typical LEO satellite passes.
- **Hydrometeor Fading**: Atmospheric rain attenuation follows ITU-R P.618-13, implementing specific attenuation $\gamma_R = kR^\alpha$ dB/km across precipitation rates from clear sky (0 mm/h) to severe convective downpours (100 mm/h).
- **Benchmark Baseline Configurations**: Three established schemes are evaluated: (M1) *Reactive Baseline* (standard 3GPP Event A3/A4 triggers with hysteresis and time-to-trigger timers); (M2) *Threshold-Based* (static 10 dB SNR margin); and (M3) *Prediction-Only* (neural point forecast without conformal uncertainty bounds).

6. Experimental Evaluation

The proposed architecture was evaluated across an exhaustive **400-trial Monte Carlo** simulation campaign under stochastic rain fading, dynamic multi-user mobility across five topological environments, and orbital satellite transitions. The reported experimental results are summarized in Table 1.

Table 1: Reported Experimental Benchmark Results (400-Trial Monte Carlo Campaign)

Orchestration Methodology	PDR (%)	99% Tail Latency	Outage Rate	Ping-Pong / hr	Energy (J/Mbit)
M1: Reactive Baseline (3GPP A3/A4)	91.20 ± 1.45	38.45 ms	8.80%	18.4 ± 2.1	4.85 ± 0.35
M2: Threshold-Based Heuristic	94.65 ± 1.10	26.12 ms	5.20%	12.1 ± 1.6	3.92 ± 0.28
M3: Prediction-Only (Point-Forecast)	97.15 ± 0.85	14.80 ms	2.95%	6.8 ± 1.1	3.15 ± 0.22
M4: CAPRO (Proposed Framework)	$\mathbf{99.41 \pm 0.32}$	**6.82** ms	**0.59%**	$\mathbf{1.9 \pm 0.4}$	$\mathbf{2.10 \pm 0.16}$
Measured Relative Improvement	**+8.21% Gain**	**-82.26% Reduction**	**-93.30% Reduction**	**-89.30% Reduction**	**-56.70% Efficiency**

Note: Reported experimental results from the verified 400-trial Monte Carlo benchmark campaign.

The reported experimental data confirms four pivotal performance findings:

1. **Packet Delivery Ratio (99.41%)**: CAPRO attains a 99.41% delivery ratio, significantly outperforming reactive baselines (91.20%) and point prediction (97.15%). By preemptively transferring traffic to stratospheric HAPS prior to convective rain cells, CAPRO maintains link continuity above 98% even during intense precipitation (100 mm/h).
2. **Deterministic Tail Latency (6.82 ms)**: CAPRO achieves a 99th-percentile tail latency of 6.82 ms, strictly satisfying the sub-10ms URLLC requirement. In contrast, reactive baselines experience large tail delays (38.45 ms) caused by reconnection and packet buffering during sudden outages.
3. **Outage Rate Suppression (0.59%)**: Network outage probability drops from 8.80% to 0.59%, representing a 93.3% relative outage reduction across diverse terrestrial clutter and weather states.
4. **Handover Stability & Energy Efficiency**: Risk-adjusted utility suppresses unnecessary handover ping-pong flapping by **89.3%** (1.9 vs. 18.4 events/hr), dramatically minimizing signaling overhead and cutting transmission energy per megabit by 56.7% (2.10 vs. 4.85 J/Mbit).

7. Research Significance and Applications

The validated algorithmic contributions provide practical operational benefits across critical 6G deployment domains:

- **Aeronautical & Maritime Routes**: Guarantees continuous, low-latency broadband for transoceanic flights and deep-sea vessels traversing satellite spot-beam boundaries outside terrestrial radar coverage.
- **Post-Disaster Communications**: Rapidly restores resilient connectivity by dynamically bridging terrestrial first responders to stratospheric HAPS and UAV relays when terrestrial cell towers are incapacitated.
- **Massive Remote IoT**: Provides deterministic telemetry collection for isolated pipeline networks, national power grids, and environmental sensor clusters without local terrestrial base stations.
- **Scalable 6G Architecture**: Demonstrates a modular paradigm separating physical resource allocation (PPO) from cross-tier proactive handover routing (CAPRO) for future mega-constellations.

8. Conclusion and Future Scope

ORBIT-INTEL 6G presents a comprehensive framework integrating physical-layer deep reinforcement learning with network-layer predictive reliability orchestration for 6G Space-Air-Ground Non-Terrestrial Networks. Combining PPO resource management with distribution-free split-conformal uncertainty calibration enables proactive circumvention of severe atmospheric rain fade and rapid satellite transitions. The 400-trial Monte Carlo evaluation verifies a 99.41% packet delivery ratio, a 6.82 ms tail latency meeting URLLC requirements, and an 89.3% reduction in handover ping-pong instability. Future work includes federated reinforcement learning across inter-satellite cross-links, software-defined radio (SDR) hardware-in-the-loop testbed validation, and dynamic spectrum co-existence with terrestrial cellular networks.

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